PCAP™ – Certified Associate in Python Programming (Exam PCAP-31-03) – EXAM SYLLABUS

PCAP-31-03 Exam



Status: Live & Active

The exam consists of five sections:

Section 1 → 6 items	Max Raw Score: 12 (12%)
Section 2 → 5 items	Max Raw Score: 14 (14%)
Section 3 → 8 items	Max Raw Score: 18 (18%)
Section 4 → 12 items	Max Raw Score: 34 (34%)
Section 5 → 9 items	Max Raw Score: 22 (22%)

Last updated: March 7, 2022

Aligned with Exam PCAP-31-03

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Section 1: Modules and Packages (12%)

Objectives covered by the block (6 exam items)

PCAP-31-03 1.1 – Import and use modules and packages

- import variants: import, from import, import as, import *
- advanced qualifying for nested modules
- the *dir()* function
- the *sys.path* variable

PCAP-31-03 1.2 – Perform evaluations using the *math* module

- functions: ceil(), floor(), trunc(), factorial(), hypot(), sqrt()

PCAP-31-03 1.3 – Generate random values using the *random* module

- functions: random(), seed(), choice(), sample()

PCAP-31-03 1.4 – Discover host platform properties using the *platform* module

- functions: platform(), machine(), processor(), system(), version(), python_implementation(), python_version_tuple()

PCAP-31-03 1.5 – Create and use user-defined modules and packages

- idea and rationale;
- the *__pycache__* directory
- the *__name__* variable
- public and private variables
- the *__init__.py* file
- searching for/through modules/packages
- nested packages vs. directory trees

Section 2: Exceptions (14%)

Objectives covered by the block (5 exam items)

PCAP-31-03 2.1 – Handle errors using Python-defined exceptions

- except, except:-except, except:-else:, except (e1, e2)
- the hierarchy of exceptions

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- raise, raise ex
- assert
- event classes
- except E as e
- the *arg* property

PCAP-31-02 2.2 – Extend the Python exceptions hierarchy with self-defined exceptions

- self-defined exceptions
- defining and using self-defined exceptions

Section 3: Strings (18%)

Objectives covered by the block (8 exam items)

PCAP-31-03 3.1 – Understand machine representation of characters

- encoding standards: ASCII, UNICODE, UTF-8, code points, escape sequences

PCAP-31-03 3.2 – Operate on strings

- functions: *ord()*, *chr()*
- indexing, slicing, immutability
- iterating through strings, concatenating, multiplying, comparing (against strings and numbers)
- operators: *in*, *not in*

PCAP-31-03 3.3 – Employ built-in string methods

- methods: *.isxxx()*, *.join()*, *.split()*, *.sort()*, *sorted()*, *.index()*, *.find()*, *.rfind()*

Section 4: Object-Oriented Programming (34%)

Objectives covered by the block (12 exam items)

PCAP-31-03 4.1 – Understand the Object-Oriented approach

- ideas and notions: class, object, property, method, encapsulation, inheritance, superclass, subclass, identifying class components

PCEP-31-03 4.2 – Employ class and object properties

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- instance vs. class variables: declarations and initializations
- the *__dict__* property (objects vs. classes)
- private components (instances vs. classes)
- name mangling

PCAP-31-03 4.3 – Equip a class with methods

- declaring and using methods
- the *self* parameter

PCAP-31-03 4.4 – Discover the class structure

- introspection and the *hasattr()* function (objects vs classes)
- properties: *__name__*, *__module__* , *__bases__*

PCAP-31-03 4.5 – Build a class hierarchy using inheritance

- single and multiple inheritance
- the *isinstance()* function
- overriding
- operators:
- not is, *is*
- polymorphism
- overriding the *__str__()* method
- diamonds

PCAP-31-03 4.6 – Construct and initialize objects

- declaring and invoking constructors

Section 5: Miscellaneous (22%)

Scope: List Comprehensions, Lambdas, Closures, and I/O Operations

Objectives covered by the block (9 exam items)

PCAP-31-03 5.1 – Build complex lists using list comprehension

- list comprehensions: the *if* operator, nested comprehensions

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PCAP-31-03 5.2 – Embed lambda functions into the code

- lambdas: defining and using lambdas
- self-defined functions taking lambdas as arguments
- functions: *map()*, *filter()*

PCAP-31-03 5.3 – Define and use closures

- closures: meaning and rationale
- defining and using closures

PCAP-31-03 5.4 – Understand basic Input/Output terminology

- I/O modes
- predefined streams
- handles vs. streams
- text vs. binary modes

PCAP-31-03 5.5 – Perform Input/Output operations

- the *open()* function
- the *errno* variable and its values
- functions: *close()*, *.read()*, *.write()*, *.readline()*, *readlines()*
- using bytearray as input/output buffer

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